04/11/2026

CODE-C
Phase-3

Aakash
Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

FORTNIGHTLY TEST FOR NEET-2027 (XII-PASSED)

MM: 720 **Test - 04** **Time: 180 Mins.**

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Answers

1. (3)	37. (2)	73. (3)	109. (3)	145. (4)
2. (1)	38. (2)	74. (2)	110. (4)	146. (1)
3. (1)	39. (1)	75. (3)	111. (2)	147. (3)
4. (4)	40. (3)	76. (2)	112. (3)	148. (1)
5. (3)	41. (2)	77. (1)	113. (2)	149. (3)
6. (3)	42. (1)	78. (1)	114. (4)	150. (4)
7. (2)	43. (4)	79. (4)	115. (2)	151. (3)
8. (3)	44. (1)	80. (2)	116. (2)	152. (1)
9. (3)	45. (3)	81. (1)	117. (4)	153. (3)
10. (3)	46. (1)	82. (3)	118. (4)	154. (4)
11. (3)	47. (2)	83. (1)	119. (2)	155. (4)
12. (3)	48. (2)	84. (1)	120. (3)	156. (3)
13. (1)	49. (1)	85. (1)	121. (2)	157. (3)
14. (2)	50. (3)	86. (1)	122. (2)	158. (4)
15. (3)	51. (2)	87. (1)	123. (3)	159. (2)
16. (2)	52. (2)	88. (3)	124. (2)	160. (3)
17. (4)	53. (1)	89. (1)	125. (3)	161. (3)
18. (4)	54. (1)	90. (1)	126. (4)	162. (2)
19. (1)	55. (1)	91. (1)	127. (4)	163. (1)
20. (2)	56. (2)	92. (2)	128. (1)	164. (4)
21. (3)	57. (4)	93. (1)	129. (4)	165. (1)
22. (2)	58. (2)	94. (1)	130. (2)	166. (2)
23. (1)	59. (1)	95. (4)	131. (3)	167. (2)
24. (4)	60. (2)	96. (4)	132. (4)	168. (2)
25. (3)	61. (2)	97. (2)	133. (3)	169. (3)
26. (4)	62. (4)	98. (2)	134. (3)	170. (4)
27. (3)	63. (3)	99. (3)	135. (1)	171. (3)
28. (2)	64. (3)	100. (3)	136. (3)	172. (4)
29. (2)	65. (3)	101. (1)	137. (3)	173. (3)
30. (4)	66. (2)	102. (3)	138. (3)	174. (1)
31. (4)	67. (2)	103. (3)	139. (2)	175. (3)
32. (2)	68. (4)	104. (2)	140. (3)	176. (2)
33. (1)	69. (4)	105. (2)	141. (1)	177. (3)
34. (3)	70. (3)	106. (2)	142. (4)	178. (3)
35. (2)	71. (3)	107. (4)	143. (3)	179. (2)
36. (2)	72. (3)	108. (2)	144. (4)	180. (2)

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Hints and Solutions

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PHYSICS

(1) **Answer :** (3)
Solution:
Using conservation of angular momentum.

(2) **Answer :** (1)
Solution:

$v_e = \sqrt{\frac{2GM}{R}}$

$\Rightarrow v_e = \sqrt{2 \times \frac{4\pi}{3} \frac{GMR^2}{\frac{4}{3}\pi R^3}}$

$v_e = \sqrt{\frac{8}{3}\pi G\rho R^2}$

$\frac{v_A}{v_B} = \sqrt{\frac{R_A^2}{R_B^2}} = 2$

(3) **Answer :** (1)
Solution:
In planetary motion
$T^2 \propto R^3$
$T \propto R^{3/2}$

(4) **Answer :** (4)
Solution:
Gain in K.E = Loss in PE
$\frac{GMm}{2R} - \frac{GMm}{3R} = \frac{GmM}{6R} = \mathrm{KE}$

(5) **Answer :** (3)
Solution:
Mass of object remains the same everywhere
Mass $m = \frac{W_{\text{surface}}}{g} = \frac{50}{10} = 5$ kg

(6) **Answer :** (3)
Solution:
$E = \frac{-GmM}{2a}$ = constant
Force and velocity are not always perpendicular in elliptical orbit.

(7) **Answer :** (2)
Solution:

$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$

$\Rightarrow \frac{W'}{W} = \frac{R^2}{(R+h)^2}$

$\Rightarrow \frac{1}{4} = \frac{R^2}{(R+h)^2} \Rightarrow \frac{R}{R+h} = \frac{1}{2}$

$\Rightarrow R + h = 2R$
$\Rightarrow h = R$

(8) **Answer :** (3)
Hint:
Gravitational force between two point masses separated by distance r, $F = \frac{Gm_1m_2}{r^2}$
Solution:
$F_{12} = \frac{Gm_1m_2}{r^2}$ and $F_{21} = \frac{Gm_1m_2}{r^2} \Rightarrow$ Required ratio $= 1:1$

(9) **Answer :** (3)

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Solution:

Maximum horizontal distance $R_{max.} = \frac{u^2}{g}$,

Where g' is acceleration due to gravity of the planet.

$\Rightarrow 4 = \frac{8 \times 8}{g'}$

$\Rightarrow g' = 16 \text{ m s}^{-2}$

(10) **Answer :** (3)
Solution:
The shell behaves like a point mass located at its center but only for points outside the shell. Due to absence of atmosphere on moon, atmospheric pressure is zero.

(11) **Answer :** (3)
Solution:
Orbital speed for circular orbit = $\sqrt{gR}$

Escape speed = $\sqrt{2gR}$

$\because \sqrt{gR} < \sqrt{\frac{23}{12}gR} < \sqrt{2gR}$

$\therefore$The body will move in an elliptical orbit.

(12) **Answer :** (3)
Hint:
Variation in value of g due to height.
Solution:

$W = mg = m \cdot \frac{GM}{R^2}$

$W' = mg' = m\frac{GM}{(R+h)^2}$

$\frac{W}{W'} = \left(\frac{R+h}{R}\right)^2$

$\frac{1}{0.64} = \left(\frac{R+h}{R}\right)^2$

$\frac{R+h}{R} = \frac{1}{0.8}$

$8R + 8h = 10R$

$h = \frac{2R}{8} = \frac{R}{4}$

(13) **Answer :** (1)
Solution:

$T = 2\pi\sqrt{\frac{a^3}{GM}}$ [for circular orbit]

where $M \rightarrow$ mass of sun

(14) **Answer :** (2)
Solution:
for, $r \leq R, g \propto r \rightarrow$ straight line variation
for, $r > R, g \propto \frac{1}{r^2}$

(15) **Answer :** (3)
Solution:

$\sqrt{\frac{GM}{R+h}} = \frac{2}{3}\sqrt{\frac{2GM}{R}}$

Or $\frac{1}{R+h} = \frac{4}{9} \times \frac{2}{R}$ or $h = \frac{R}{8}$

(16) **Answer :** (2)
Solution:
Using, $g_{eff} = g - \omega^2 R\cos^2\lambda$
For equator
$\lambda \rightarrow 0$
$g_{eq} = g - \omega^2 R$
If earth stops $\omega = 0$
$\therefore g_{eq} = g$ *i.e.,* increases by amount $\omega^2 R$
For pole

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$\lambda \to 90^\circ$
cos90°
$\Rightarrow g_p = g$
Hence *g* remains same if earth stops rotating

(17) **Answer :** (4)
Solution:

Energy stored = $\frac{1}{2} \times \frac{(\text{stress})^2}{Y} \times$ Volume

$U = \frac{1}{2}\left(\frac{2g}{A}\right)^2 \frac{\text{Volume}}{Y}$

$U' = \frac{1}{2}\left(\frac{4g}{A}\right)^2 \frac{\text{Volume}}{Y}$

$U' = 4U$

(18) **Answer :** (4)
Solution:
Breaking stress = $\frac{F}{A} = \frac{mg}{A}$

$\sigma = \frac{ALdg}{A}$

$\sigma = Ldg$

$\therefore L = \frac{6.4 \times 10^5}{3.2 \times 10^3 \times 10} = \frac{64}{3.2} = 20$ m

(19) **Answer :** (1)
Solution:
$V = Al,\ Y = \frac{Fl}{A\Delta l} \Rightarrow \Delta t = \frac{Fl}{AY} = \frac{Ft^2}{VY}$
$\Rightarrow \Delta l \propto l^2$

(20) **Answer :** (2)
Solution:
$\frac{Y_A}{Y_B} = \frac{\tan 53^\circ}{\tan 37^\circ} = \frac{16}{9}$
$9Y_A = 16\,Y_B$

(21) **Answer :** (3)
Solution:
$\frac{\Delta\rho}{\rho} = -\frac{\Delta V}{V} = \frac{\Delta P}{B}$
$\Delta P = 3 \times 10^9 \times \frac{0.2}{100} = 6 \times 10^6$ N/m^2

(22) **Answer :** (2)
Solution:
$B = \frac{\Delta P}{\frac{-\Delta V}{V}} = \frac{\rho g h}{\left(\frac{-\Delta V}{V}\right)}$
$B = \frac{2\times10^3\times10^3\times10}{4\times10^{-4}} = 5 \times 10^{10}\text{N/m}^2$

(23) **Answer :** (1)
Solution:
Young's modulus for Rubber is non zero.

(24) **Answer :** (4)
Solution:
Cutting the substance into pieces does not change the internal structure of a substance, hence it does not affect the elasticity.

(25) **Answer :** (3)
Solution:
Tensile stress is developed only in solids.

(26) **Answer :** (4)
Solution:
Wire 1 : $Y_1 = \frac{4g\times3}{A\times\Delta l_1}$... (i)

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Wire 2 : $Y_2 = \frac{3g\times 1}{A\times\Delta l_2}$... (ii)

Divide eq. (i) by (ii)

$\frac{Y_1}{Y_2} = \frac{4g\times 3}{A\times\Delta l_1} \times \frac{A\times\Delta l_2}{3g\times 1}$

$\frac{Y_1}{Y_2} = \frac{4\Delta_2}{\Delta l_1}$

$\frac{\Delta l_1}{\Delta l_2} = \frac{4Y_2}{Y_1}$

= 4 × 3 : 1

= 12 : 1

(27) Answer : (3)

Solution:

$B = -\frac{P}{\frac{\Delta V}{V}}$

$\frac{\Delta V}{V} = \frac{P}{B}$

$\therefore \frac{\Delta r}{r} = \frac{P}{3B}$

(28) Answer : (2)

Solution:

Breaking stress remains constant. Breaking stress is a characteristic property of the material of rope.

(29) Answer : (2)

Solution:

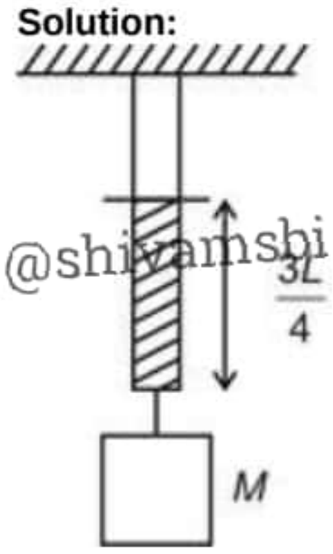
Hooke's law is obeyed within proportional limit.

(30) Answer : (4)

Hint:

Stress will be different at different points of wire.

Solution:



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Tension at $\frac{3L}{4}$ from lower end

$T = Mg + \frac{3Mg}{4}$

$= 7\frac{Mg}{4}$

Stress developed at this point

$S = \frac{F}{A} = \frac{7Mg}{4A}$

(31) Answer : (4)

Solution:

$3P = P + hpg$ ...(i)

$2P = hpg$

$P' = P + \frac{3}{5}h\rho g$

$P' = P + \frac{3}{5}(2P) = P + \frac{6P}{5} = \frac{11}{5}P$

(32) Answer : (2)

Solution:

$\frac{V_0}{V}\times 100 = \left(1-\frac{\rho}{\sigma}\right)\times 100$

$= \left(1-\frac{480}{800}\right)\times 100$

= 40%

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(33) **Answer :** (1)
Solution:
$V_t = \frac{2}{9\eta} r^2 \left(\rho - \sigma\right) g$

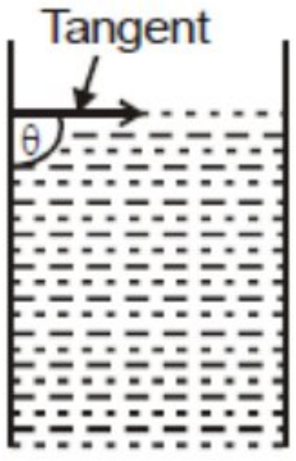
(34) **Answer :** (3)
Solution:
Bernoulli's equation is based upon energy conservation.

(35) **Answer :** (2)
Solution:
By Bernouli's equation.
$v = \sqrt{2gh_2}$

(36) **Answer :** (2)
Solution:
Surface tension energy = T(2A)
U = 2 × 5 × 0.02
= 10 × 0.02
= 0.2
= 2 × 10^{-1} J

(37) **Answer :** (2)
Solution:
Cohesive forces between molecules.

(38) **Answer :** (2)
Solution:
The angle between the tangent to the liquid surfaceand the solid surface inside the liquid is θ = 90°.



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(39) **Answer :** (1)
Solution:
$\frac{4}{3}\pi R^3 = 8 \times \frac{4}{3}\pi r^3$
$\therefore\ R = 2r$
$\Delta A = 8 \times 4\pi r^2 - 4\pi R^2 = 4\pi \times 10^{-6}$
$Energy = \Delta A \times S.T. = 9 \times 10^{-7} J$

(40) **Answer :** (3)
Solution:
$\frac{2T}{r} = h\rho g$
$\Rightarrow h = \frac{2T}{r\rho g}$

(41) **Answer :** (2)
Solution:
Rate of flow = $A\sqrt{2gh}$
$= \left(10^{-3}\right)^2 \sqrt{2 \times 10 \times 0.2}$ m^3/s
2 × 10^{-6} × 10^6 cm^3/s
= 2 cm^3/s

(42) **Answer :** (1)
Solution:
$A_1 v_1 = A_2 v_2$

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$\pi\frac{16}{4} \times 0.5 = \pi \times \frac{1}{4} \times v$

$\therefore v = 8$

Range = $v \times t = 8 \times \sqrt{\frac{2h}{g}}$ = 4 m

(43) **Answer :** (4)
Solution:
Initially velocity increases and after some time body attains terminal velocity.

(44) **Answer :** (1)
Solution:
Use, $h = \frac{2T\cos\theta}{r\rho g}$

$\theta < 90°$	$\cos\theta = +$	Concave meniscus
$\theta > 90°$	$\cos\theta = (-)$	Fall in capillary with convex meniscus
$\theta = 90°$	$\cos 90° = 0$	No rise or fall in capillary

(45) **Answer :** (3)
Solution:
The breaking stress of a wire depends on nature of the material of the wire. For an ideal liquid, the shear modulus is zero and the bulk modulus is infinite. Shear modulus is generally less than Young's modulus. For most materials $G \approx Y/3$

CHEMISTRY

(46) **Answer :** (1)
Hint:
Common ion effect

$M(OH)_2 + H_2O(l) \rightleftharpoons M^{2+}(aq) + 2OH^-(aq)$

At initial	0	0.2
At equilibrium state	s	$0.2+2s \approx 0.2$

$K_{sp} = [M^{2+}][OH^-]^2$

$4.0 \times 10^{-15} = (s)(0.2)^2$

$s = \frac{4\times10^{-15}}{4\times10^{-2}}$

$= 10^{-13}$ mol/L

(47) **Answer :** (2)
Hint:
Basic buffer solution ($NH_4OH + NH_4Cl$)

$pOH = pK_b + \log\frac{[Salt]}{[Base]}$

Solution:

NH_4OH	+ HCl	$\rightleftharpoons$	NH_4Cl	+ H_2O
50×0.1	25×0.1		0	0
5	2.5		0	0
$5-2.5$	$2.5-2.5$		2.5	2.5
2.5	0		2.5	2.5

$pOH = 4.74 + \log\frac{\left[\frac{2.5}{75}\right]}{\left[\frac{2.5}{75}\right]}$

pOH = 4.74
pH = 14 – 4.74 = 9.26

(48) **Answer :** (2)
Hint:

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$$A(s) \rightarrow \underset{p_1}{B(g)} + \underset{p_1+p_2}{C(g)}$$

$$D(s) \rightarrow \underset{p_1+p_2}{C(g)} + \underset{p_2}{E(g)}$$

Solution:

K_{p_1} = (p_1) $(p_1 + p_2)$

$a = p_1^2 + p_1p_2$ ...(i)

K_{p_2} = (p_2) $(p_1 + p_2)$

$b = p_2^2 + p_1p_2$ ...(ii)

$a + b = p_1^2 + 2p_1p_2 + p_2^2$

$(a + b) = (p_1 + p_2)^2$ ...(iii)

Calculating total pressure:

$p_T = p_B + p_C + p_E$

$= p_1 + p_1 + p_2 + p_2$

$= 2p_1 + 2p_2$

$= 2(p_1 + p_2)$

Putting the value of $p_1 + p_2$ in equation (iii)

$p_T = 2\left(\sqrt{a+b}\right) atm$

(49) Answer : (1)

Hint:

Removal of $SCN^-(aq)$ ion shifts the equilibrium from right to left.

Solution:

Hg^{2+} reacts with $SCN^-(aq)$ ions to form a stable complex $[Hg(SCN)_4]^{2-}(aq)$.

Removal of free $SCN^-(aq)$ ions shifts equilibrium backward and intensity of red colour due to $[Fe(SCN)]^{2+}$ decreases.

(50) Answer : (3)

Solution:

$BaSO_4 : s^2 = K_{sp}$

$\therefore \ s = \sqrt{K_{sp}}$

$Hg_2Cl_2 : s \times (2s)^2 = K_{sp}$

$= 4s^3 = K_{sp}$

$s = \sqrt[3]{\frac{K_{sp}}{4}} = \left(\frac{K_{sp}}{4}\right)^{1/3}$

$Ca_3(PO_4)_2 : (3s)^3 \times (2s)^2 = K_{sp}$

$\Rightarrow 108\, s^5 = K_{sp}$

$s = \sqrt[5]{\frac{K_{sp}}{108}}$

$Fe(OH)_3 : s \times (3s)^3 = K_{sp}$

$\Rightarrow 27s^4 = K_{sp}$

$\Rightarrow s = \sqrt[4]{\frac{K_{sp}}{27}}$

(51) Answer : (2)

Hint:

Value of K_C depends upon concentration and stoichiometric constant both.

Solution:

For very small values of K_C, reactant will dominate the equilibrium mixture.

(52) Answer : (2)

Solution:

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Catalyst does not affect the equilibrium composition of the reaction mixture.

(53) **Answer :** (1)
Hint:
At equilibrium ΔG = 0
$\Delta G = \Delta G^\circ + RT \ln k_{eq}$
Solution:

	$2HI(g) \longrightarrow$	$H_2(g)$ +	$I_2(g)$
t = 0	1 mole	0	0
	1 – 0.5	0.25	0.25

$K_c = \frac{0.25 \times 0.25}{0.5 \times 0.5} = \frac{1}{4}$
$\Delta G^\circ = -2.303\ RT \log\left(\frac{1}{4}\right)$
= 1.386 RT

(54) **Answer :** (1)
Hint:
Mixture of weak acid and its salt with strong base form an acidic buffer.

(55) **Answer :** (1)
Solution:

- K_a of $HNO_2 = 4 \times 10^{-4}$
  $pK_a = -\log\left(4 \times 10^{-4}\right)$
  $= 4 - \log 4 = 4 - 2 \times 0.3 = 4 - 0.6 = 3.4$
  $pK_a\left(HNO_2\right) + pK_b\left(NO_2^-\right) = 14$
  $3.4 + pK_b = 14$
  $pK_b = 14 - 3.4 = 10.6$

(56) **Answer :** (2)
Hint:
$K_p = \frac{P_{CO_2}}{P_{CO}}$
P = equilibrium partial pressure
Solution:

	$FeO(s) + CO(g) \rightleftharpoons$	$Fe(s) + CO_2(g)$
initial pressure	4	0
At equilibrium	4 – x	x

$K_p = \frac{P_{CO_2}}{P_{CO}}$
$K_p = \frac{x}{4-x}$
$1.5 = \frac{x}{4-x}$
x = 6 – 1.5x
$x = \frac{6}{2.5} = 2.4$
Equilibrium partial pressure of CO(g)
= 4 – 2.4 = 1.6 atm

(57) **Answer :** (4)
Solution:
$XeF_6(g) + H_2O(g) \rightleftharpoons XeOF_4(g) + 2HF(g)\ (K_1)$ ...(i)
$XeO_4(g) + XeF_6(g) \rightleftharpoons XeOF_4(g) + XeO_3F_2(g)\ (K_2)$ ...(ii)
Required equation : (ii) – (i)
$K = \frac{K_2}{K_1}$

(58) **Answer :** (2)
Hint:
For any gaseous equilibrium
$K_P = K_C(RT)^{\Delta ng}$
Solution:
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $K_P = K_C$

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$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $K_P = K_C(RT)^{-2}$

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_P = K_C(RT)$

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $K_P = K_C(RT)^{-1}$

(59) **Answer :** (1)
Hint:
Electron deficient species act as Lewis acid.
Solution:
$AlCl_3$ behaves as Lewis acid.

(60) **Answer :** (2)
Hint:
$M_1V_1 + M_2V_2 = M_RV_R$
Solution:

$pH_1 = 2$	$pH_2 = 3$
$M_1 = [H^+]_1 = 10^{-2}$	$[H^+]_2 = 10^{-3} = M_2$
$V_1 = 100$ ml	$V_2 = 100$ ml

$M_R = \frac{10^{-2}\times 100 + 10^{-3}\times 100}{200} = 5.5 \times 10^{-3}$

$pH_{(Resultant\ solution)} = -\log[5.5 \times 10^{-3}]$

$pH_{(R)} = 2.26$

(61) **Answer :** (2)
Hint:
$pH = \frac{1}{2}[pK_w + pK_a + \log C]$
Solution:
For salt of weak acid and strong base

$pH = \frac{1}{2}[pK_w + pK_a + \log C]$

$pH = \frac{1}{2}[14 + 4.76 + \log 0.1]$

$= \frac{1}{2}[18.76 - 1]$

$= \frac{17.76}{2} = 8.88 \approx 8.9$

(62) **Answer :** (4)
Solution:
Equilibrium constant is a function of temperature only.

(63) **Answer :** (3)
Hint:
Exothermic reactions are favoured at low temperature.
Solution:
Addition of inert gas at constant volume has no effect on equilibrium state.
When we decrease the pressure, reaction will shifts towards that side which has higher number of gaseous moles.

(64) **Answer :** (3)
Solution:

When $\left[\overline{O}\,H\right]_{SB} < 10^{-7}M$, contribution of OH^- from H_2O is also considered

$H_2O \rightleftharpoons \underset{x}{H^+}(aq) + \underset{10^{-8}+x}{OH^-}(aq)$

$10^{-14} = x\left(10^{-8} + x\right)$

Solving quadratic,

$[H^+] = x = 9.5 \times 10^{-8}$

pH = 7.02

(65) **Answer :** (3)
Hint:
Algebraic summation of oxidation state of each element in a charged species is equal to its charge.
Solution:
• Oxidation state of Cl in $NaClO_3$ is +5 but highest oxidation state of Cl is +7.

• There is no oxidation state change during conversion of $Cr_2O_7^{2-}$ to CrO_4^{2-} hence it does not require any OA or RA.

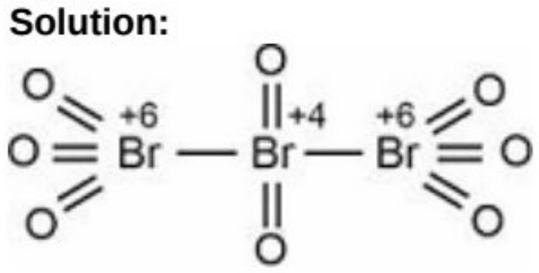
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• Oxidation state of Cl in $CaOCl_2$ is +1 and –1.

(66) **Answer :** (2)
Solution:
On balancing the given redox reaction:
$2\,MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2\,Mn^{2+} + 10\,CO_2 + 8H_2O$
5 mol of $C_2O_4^{2-} \equiv$2 mol of MnO_4^-
$\therefore$ 0.5 mol of $C_2O_4^{2-} \equiv$0.2 mol of MnO_4^-

(67) **Answer :** (2)
Hint:
For covalent bond, high electronegative element gets –1 oxidation state and less electronegative element gets +1 oxidation state
Solution:

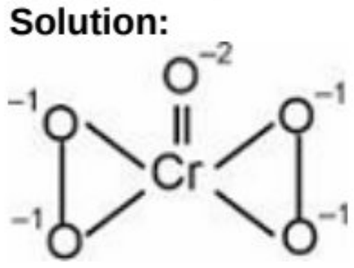


Average oxidation state of $Br = \frac{+16}{3}$

(68) **Answer :** (4)
Hint:
For a compound, net charge is zero.
Solution:
A_2BC_2 : Net charge = 2 × (1) + 4 + 2 × (–3) = 0

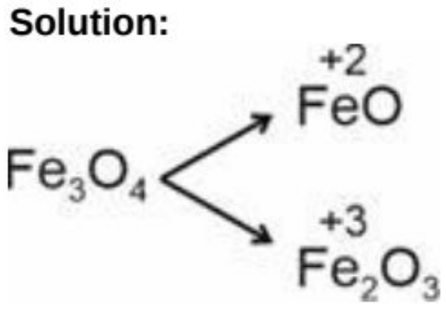
(69) **Answer :** (4)
Hint:
Oxidation states of some elements changes during a redox reaction.
Solution:
$KClO_3 \xrightarrow{\Delta} KCl + O_2$; Decomposition redox
Zn + $CuSO_4$ → $ZnSO_4$ + Cu; Metal displacement redox
Na + H_2O → NaOH + H_2; Non-metal displacement redox
$CaCO_3$ → CaO + CO_2 ; Non-redox

(70) **Answer :** (3)
Hint:
CrO_5 has 2 peroxide linkages.
Solution:



Oxidation state of Cr be x
$\therefore$ x + (–1) × 4 + (–2) = 0
$\Rightarrow$ x = +6

(71) **Answer :** (3)
Hint:
Fe_3O_4 is a mixed oxide
Solution:



(72) **Answer :** (3)
Solution:
Greater is the standard reduction potential, lesser is the reducing power.

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Na
Mg
Al
Zn

$E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$ $E^{\circ}_{Al^{3+}/Al} = -1.66\ V$

$E^{\circ}_{Mg^{2+}/Mg} = -2.36\ V$ $E^{\circ}_{Na^{+}/Na} = -2.71\ V$

Order of reducing power
Na > Mg > Al > Zn

(73) **Answer :** (3)
Solution:
E_{cell} is an intensive property and it does not depend on the number of electrons lost or gained during the process.
According to the equation,
$\Delta G = -nFE_{cell}$
The value of ΔG depends on n value.

(74) **Answer :** (2)
Hint:
$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log Q$
Solution:

$$A(s) + 2B^{+}_{0.1\,M} \rightleftharpoons A^{2+}_{0.1\,M} + 2B(s)$$

$Q = \frac{[A^{2+}]}{[B^{+}]^2} = \frac{(0.1)}{(0.1)^2} = 10$

$5 = E^{\circ}_{cell} - \frac{0.0591}{2} \log 10$
$E^{\circ}_{cell} = 5.03\ V$

(75) **Answer :** (3)
Hint:
$Al^{3+} + 3e^{-} \rightarrow Al$
1 mole = 27g
Solution:
2F will deposite $\frac{27\times2}{3}$ g = 18 g Al

(76) **Answer :** (2)
Hint:
$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$.
Solution:
$E^{\circ}_{cell} = E^{\circ}_{Hg^{2+}/Hg_2^{2+}} - E^{\circ}_{Pb^{2+}/Pb}$
= 0.97 – (– 0.13) = 1.1 V

(77) **Answer :** (1)
Hint:
$2H^{+}(aq) + 2e^{-} \rightarrow H_2(g)$
$E_{H^+/H_2} = E^{\circ}_{H^+/H_2} - \frac{0.0591}{2} \log \frac{pH_2}{[H^+]^2}$
Solution:
$E_{H^+/H_2} = 0 - \frac{0.0591}{2} \log \frac{1}{(10^{-5})^2}$
$= -\frac{0.0591}{2} \log 10^{10}$
$= -\frac{0.0591}{2} \times 10 = -0.296\ V$

(78) **Answer :** (1)
Hint:
Cell constant = $\left(\frac{1}{a}\right)$
Solution:
Unit of cell constant is cm^{-1} or m^{-1}.

(79) **Answer :** (4)

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Solution:

Batteries/Cells	Electrolytes
Dry cell	Moist paste of NH_4Cl and $ZnCl_2$
Mercury cell	Paste of KOH and ZnO
(H_2–O_2) Fuel cell	Conc. (aq) NaOH solution
Lead storage battery	38% solution of H_2SO_4

(80) **Answer :** (2)
Hint:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

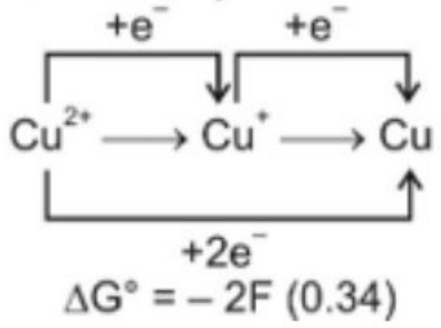
Solution:
n-factor = 5
Number of Faraday required = 0.5 × 5 = 2.5 F
Electricity required = 2.5 × 96500 C

(81) **Answer :** (1)
Hint:

$Cu^{2+} \rightarrow Cu^{+} \rightarrow Cu$

Solution:

$\Delta G^\circ = -1Fx,\ \Delta G^\circ = -1F\ (0.52)$



$\Delta G^\circ = -2F\ (0.34)$

– 1F(x) – 1F(0.52) = – 2F(0.34)
⇒ x = 0.68 – 0.52 = 0.16 V

(82) **Answer :** (3)
Solution:
Mercury cell:
Anode: Paste of Zn-Hg amalgam
Cathode: Paste of HgO + C

(83) **Answer :** (1)
Solution:

$2Fe\,(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O\,(l)$

Anode: $2Fe(s) \rightarrow 2Fe^{2+}(aq) + 4e^-$

Cathode: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

(84) **Answer :** (1)
Solution:

$$NaCl\left(aq.\right) \xrightarrow{\text{Electrolysis}} \begin{matrix} H_2(g) + Cl_2(g) \\ \text{dilute solution} + NaOH(aq.) \end{matrix}$$

(85) **Answer :** (1)
Solution:

Molar conductivity (Λ_m) $= \frac{k \times 1000}{C}$

$\Lambda_m = \frac{0.015 \times 1000}{0.2} = 75\ S\ cm^2\ mol^{-1}$

(86) **Answer :** (1)
Solution:

$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{20}{600} = 0.033$

As α = 1

So, $K_a = C\alpha^2 = \frac{1}{10} \times 0.0333 \times 0.0333$ $= 1.1 \times 10^{-4}$

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(87) **Answer :** (1)
Solution:

$\frac{w_{Ag}}{w_{Cu}} = \frac{E_{Ag}}{E_{Cu}} \Rightarrow w_{Cu} = \frac{0.54 \times 63.5}{108 \times 2} \approx 0.158$ g

(88) **Answer :** (3)
Solution:
According to Kohlrausch law of independent migration of ions.
$\Lambda_m^o(CH_3COOH) = \Lambda_m^o(CH_3COONa) + \Lambda_m^o(HCl) - \Lambda_m^o(NaCl)$
= 91.0 S cm^2 mol^{-1} + 426.16 S cm^2 mol^{-1}
– 126.45 S cm^2 mol^{-1}
= 390.71 S cm^2 mol^{-1}

(89) **Answer :** (1)
Solution: Except conductivity(k), all will be increasing on dilution

(90) **Answer :** (1)
Solution:

Ions	H^+	Na^+	Mg^{2+}	Ca^{2+}
λ° (S cm^2 mol^{-1})	349.6	50.1	106.0	119.0

BOTANY

(91) **Answer :** (1)
Solution:
The trichomes in the shoot system are usually multicellular. They may be branched or unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

(92) **Answer :** (2)
Solution:
Guard cells are bordered by one or more modified epidermal cells called subsidiary cells or accessory cells.

(93) **Answer :** (1)
Hint:
Root hair arise from the outermost layer of root.
Solution:
Epiblema, the outermost layer of root, forms root hairs.

(94) **Answer :** (1)
Solution:
In stem and leaves of monocots and in dicot leaves, vascular bundles are conjoint and closed. In dicot stem,conjoint and open type of vascular bundle is found.

(95) **Answer :** (4)
Hint:
Ground tissue system constitutes the main bulk of plant body.
Solution:
All tissues except epidermis and vascular bundles constitute the ground tissue.

(96) **Answer :** (4)
Solution:
Stomatal apparatus consist of stomatal pore, guard cells and surrounding subsidiary cell.

(97) **Answer :** (2)
Solution:
Hypodermis is collenchymatous in dicot stem.

(98) **Answer :** (2)
Solution:

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In dicot roots, xylem patches are two to four in number.

(99) **Answer :** (3)
Solution:
The type of vascular bundle in which cambium is present is called open.

(100) **Answer :** (3)
Hint:
Lateral roots originate from the outermost part of the stele.
Solution:
In the dicot root, the lateral roots originates from the pericycle.

(101) **Answer :** (1)
Solution:
Cortex is the region present between epidermis and stele.

(102) **Answer :** (3)
Solution:
Hint: Conjunctive tissue is present in between the patches of xylem and phloem.
Sol.: The parenchymatous cells which lie between the xylem and the phloem of both monocotyledonous and dicotyledonous roots are called conjunctive tissue.

(103) **Answer :** (3)
Solution:
Hypodermis of dicot stem consists of collenchymatous cells but cortical layers below hypodermis consist of thin walled parenchymatous cells with conspicuous intercellular spaces. Casparian strips are found in the endodermis, which is not a part of stele region.

(104) **Answer :** (2)
Solution:
Sol.: The cells of the endodermis are rich in starch grains, and is known as starch sheath. It is found in dicot stems.

(105) **Answer :** (2)
Solution:
Water-containing cavities are present within the vascular bundles of monocot stem.

(106) **Answer :** (2)
Solution:
Hypodermis is the outermost sub-zone of cortex in stems and is absent in roots.

(107) **Answer :** (4)
Hint:
These cells help in folding of leaves during water stress.
Solution:
Bulliform cells are large, empty, colourless cells of the adaxial epidermis along the veins of grass leaves.

(108) **Answer :** (2)
Solution:
Bulliform cells are observed in monocot leaf not in dicot leaf.

(109) **Answer :** (3)
Solution:
In monocot stem, vascular bundles are conjoint and closed. These are scattered and surrounded by bundle sheath cells.

(110) **Answer :** (4)
Solution:
Mesophyll of dicot leaves is differentiated into palisade and spongy parenchyma.

(111) **Answer :** (2)
Hint:
Large and colourless cells are present in adaxial epidermal cells of grass leaves (monocot).
Solution:
Bulliform cells in monocot become flaccid due to water stress and make the grass leaves curl inwards to minimise water loss.

(112) **Answer :** (3)
Solution:
Classification for flowering plants given by George Bentham and Joseph Dalton Hooker is also called natural classification system. This system is based on natural affinities among the organisms and considers not only external features, but also internal features like ultrastructure, anatomy, embryology, etc.

(113) **Answer :** (2)

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Solution:
Cytotaxonomy is based on chromosome number, structure and behavior.

(114) **Answer :** (4)
Solution:
Certain marine brown and red algae produce large amounts of hydrocolloids. *e.g.*: algin (brown algae) and carrageen (red algae).

(115) **Answer :** (2)
Hint:
Union of two gametes is called syngamy.
Solution:
In anisogamy, two kinds of distinct gametes are formed. In oogamy, one motile male gamete and other non-motile female gamete is formed.

(116) **Answer :** (2)
Solution:
Atleast a half of the total CO_2 fixation on the earth is carried out by algae through photosynthesis.

(117) **Answer :** (4)
Solution:
In gymnosperms, the transfer of pollen grains occur by air current.

(118) **Answer :** (4)
Hint:
The alga shown in the diagram is *Laminaria*.
Solution:
Laminaria belongs to Phaeophyceae.
It stores food in the form of laminarin.
It is rich in iodine and is edible.
It shows dominance of fucoxanthin.

(119) **Answer :** (2)
Solution:
Haploid sex organs are present on gametophyte in bryophytes.

(120) **Answer :** (3)
Hint:
In brown algae, the union of gametes may take place in water or within the oogonium.
Solution:
Sargassum is the brown alga in which sexual reproduction takes place by fusion of an egg and motile male gamete.

(121) **Answer :** (2)
Solution:
Liverworts produce motile male gametes which transfer to female sex organ via water.

(122) **Answer :** (2)
Solution:
In bryophytes, zygote undergoes equational division to form sporophyte and spore mother cell undergoes meiosis to form haploid spores.

(123) **Answer :** (3)
Solution:
- *Volvox* is a colonial alga . *Funaria* being a moss has elaborate mechanism of spore dispersal. *Gracilaria* is a red alga. Red algae reproduce sexually by non-motile gametes. Pteridophytes like *Equisetum* are typically homosporous and bear sporangia in distinct compact structures called strobili or cones.

(124) **Answer :** (2)
Solution:
In pteridophyte, the embryo formed after fusion of gamete develops into a multicellular well differentiated sporophyte.

(125) **Answer :** (3)
Hint:
Integument is the protective covering of ovule.
Solution:
Nucellus is protected by envelopes and the composite structure is called an ovule.

(126) **Answer :** (4)
Hint:
Ginkgo is a gymnosperm.

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Solution:
Adiantum is a pteridophyte.

(127) **Answer :** (4)
Solution:
The gymnosperms are heterosporous; they produce haploid microspores and megaspores.

(128) **Answer :** (1)
Solution:
Sequoia is one of the tallest gymnosperms.

(129) **Answer :** (4)
Hint:
Compact, spirally arranged sporophylls are called strobili or cone.
Solution:
Equisetum, *Selaginella* and *Pinus* bear strobili or cone. Cones are not found in bryophytes.

(130) **Answer :** (2)
Hint:
In *Cycas,* megasporophylls do not aggregate compactly.
Solution:
Cycas lacks female cones.

(131) **Answer :** (3)
Hint:
Eudorina shows anisogamy.
Solution:
Algae has oogonium as sex organ and archegonium is absent in them. So Eudorina has oogonium.

(132) **Answer :** (4)
Solution:
Leaves of conifers have thick cuticle, sunken stomata and are needle shaped to withstand extreme conditions and to prevent water loss.

(133) **Answer :** (3)
Solution:
Parenchyma cells has thin non lignified cell wall composed of cellulose mainly.

(134) **Answer :** (3)
Solution:
Pteridophytes have sporophyte as main dominant body. They have both sporophyte as well as gametophytic bodies independent.

(135) **Answer :** (1)
Hint:
Members of Rhodophyceae lack flagella.
Solution:
All three classes of algae exhibit fragmentation as the mode of vegetative reproduction.

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ZOOLOGY

(136) **Answer :** (3)
Solution:
Sweat contains, water, NaCl, urea, lactic acid, *etc.,* but not waxes.

(137) **Answer :** (3)
Solution:
Podocytes are arranged in an intricate manner to leave some minute spaces called slit pores. Blood is filtered finely through these membranes, that almost all the constituents of the plasma, except the proteins, pass into the lumen of the Bowman's capsule.

(138) **Answer :** (3)
Solution:
Initial filtrate osmolarity is 300 mOsmol L^{-1} in the cortex and reaches 1200 mOsmol L^{-1} in the inner medulla. Human kidneys can produce urine nearly four times concentrated than initial filtrate.

(139) **Answer :** (2)

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Solution:
Small amounts of urea enters the thin segment of the ascending limb of Henle's loop.
Collecting duct allows the passage of small amount of urea into medullary interstitium to maintain osmolarity, which is picked up by thin segment of ascending limb of Henle's loop. The counter current mechanism helps to maintain a concentration gradient in the medullary interstitium.

(140) Answer : (3)
Hint:
CO = 5040 ml
Solution:

On an average, 1100-1200 mL of blood is filtered by the kidneys per minute which constitutes roughly 1/5th of the blood pumped out by each ventricle of heart in a minute.

(141) Answer : (1)
Solution:
Extensions of cortex between medullary pyramids of kidney are known as Columns of Bertini.

(142) Answer : (4)
Hint:
A nitrogenous waste
Solution:
The tubular epithelial cells in different segments of nephron perform reabsorption by either active or passive mechanisms. Substances like glucose, amino acids, Na^+, *etc.,* in the filtrate are reabsorbed actively whereas nitrogenous wastes are absorbed by the passive transport.

(143) Answer : (3)
Solution:
Hint: About 95% of urine is water
Sol.: An average adult excretes about 1 to 1.5 L of urine/day. Which contains about 25-30 g urea.

(144) Answer : (4)
Hint:
Hair pin like structure.
Solution:
Ascending limb of loop of Henle is impermeable to water but permeable to salts. Reabsorption of water occurs at PCT, DCT, CD and descending limb of loop of Henle.

(145) Answer : (4)
Solution:
(A) Mammals - Ureotelic
(B) Reptiles and lands snails - Uricotelic

(146) Answer : (1)
Solution:
70 to 80% of water is reabsorbed in PCT.

(147) Answer : (3)
Solution:
Anticoagulant heparin is added to the blood before pumping it into dialysing unit.
Histamine is released by mast cell during allergic conditions.
Serotonin is a potent vasoconstrictor.-

(148) Answer : (1)
Solution:
ANF causes vasodilation (dilation of blood vessel) and thereby decreases the blood pressure.

(149) Answer : (3)
Solution:
From the collecting duct, the filtrate reaches renal calyx and then through renal pelvis, it is drained to the ureter and finally to the urinary bladder for storage and micturition.

(150) Answer : (4)
Hint:
Identify a free living flatworm.
Solution:
Protonephridia or flame cells are the excretory structures in Platyhelminthes, rotifers, some annelids and cephalochordates.

(151) Answer : (3)
Hint:

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Ascending limb of loop of Henle is permeable to salt.
Solution:
Descending limb is called concentrating segment of loop of Henle as it is permeable to water and almost impermeable to salts.

(152) **Answer :** (1)
Solution:
Kidneys are located between the levels of last thoracic and third lumbar vertebrae.

(153) **Answer :** (3)
Hint:
Approximately 200 mL of CO_2 is removed per minute
Solution:
Our lungs remove approximately 200 mL of CO_2 in one minute.
1 hour = 60 minutes
Hence, in 60 minutes our lungs will remove 200 mL × 60 minutes = 12000 mL/hour.

(154) **Answer :** (4)
Hint:
This promotes formation of red blood cells.
Solution:
Erythropoietin released by JG cells of kidney stimulates formation of RBCs. Aldosterone targets DCT of nephrons but is produced by adrenal cortex. Rennin is a digestive enzyme.

(155) **Answer :** (4)
Solution:
Anti-diuretic hormone is secreted by hypothalamus and stored in posterior pituitary. It stimulates reabsorption of water and electrolytes by the distal tubules and reduces loss of water through urine.

(156) **Answer :** (3)
Solution:
During muscle contraction, actin filaments slide over the myosin filaments which results in shortening of muscle. Myosin constitutes the thick filament and is a polymer of meromyosin.
Collagen fibres are proteinaceous fibres found in connective tissues.

(157) **Answer :** (3)
Solution:
Some specialised cells in our body like macrophages and leucocytes exhibit amoeboid movement.

(158) **Answer :** (4)
Solution:
Troponin masks the active binding sites on the actin filaments.

(159) **Answer :** (2)
Hint:
Similar to what happens to 'I' bands during muscle contraction
Solution:
The 'Z' line attached to actins are pulled inwards, thereby causing a shortening of sarcomere, *i.e.*, contraction. During contraction, the 'I' bands get reduced, whereas the 'A' bands retain its length. The length of 'H' zone also decreases during muscle contraction. Length of both thin and thick filaments remain unchanged.

(160) **Answer :** (3)
Solution:
Acetylcholine is the specific neurotransmitter released at the neuromuscular junction. When it binds to receptors on the sarcolemma, it triggers an action potential in the muscle fibre.

(161) **Answer :** (3)
Solution:
Contractile proteins (Generate force during muscle contraction)– Actin and myosin.
Regulatory proteins(Proteins that help to switch muscle contraction process on and off) – Tropomyosin and troponin.
Structural proteins (Keep thick and thin filaments in proper alignment) – Dystrophin.

(162) **Answer :** (2)
Solution:
Endoplasmic reticulum also known as sarcoplasmic reticulum is a rich source of calcium ions, in a muscle fibre.

(163) **Answer :** (1)
Hint:
Muscular movement is performed by muscles.

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Solution:
Cells of the human body exhibit three main types of movement, namely amoeboid, ciliary and muscular. Both neural and muscle fibres show excitability but contractility is shown only by muscle fibres.

(164) **Answer :** (4)
Hint:
Their activity can be regulated by our will.
Solution:
Thigh muscle is an example of skeletal muscle which is characterised by the muscle fibres that are striated, voluntary in nature and multinucleated.

(165) **Answer :** (1)
Hint:
Also shown by neutrophils.
Solution:
Some specialised cells in our body like macrophages in tissues and leucocytes in blood exhibit amoeboid movement.
Cytoskeletal elements like microfilaments are also involved in amoeboid movement.
Our tongue shows muscular movement.
Human sperms show flagellar movement.
Transport of ovum through the fallopian tube occurs *via* ciliary movement.

(166) **Answer :** (2)
Solution:
Troponin is a regulatory protein, which is also globular and trimeric.

(167) **Answer :** (2)
Hint:
Such lines never meet.
Solution:
A characteristic feature of muscle fibre is the presence of large number of parallelly arranged filaments in the sarcaplasm called myofibrils. These myofibrils consist of two types of myofilaments actin and myosin. Each muscle bundle contains a number of muscle fibres.

(168) **Answer :** (2)
Hint:
Anaerobic breakdown of glycogen.
Solution:
During strenuous exercise, glycogen is anaerobically metabolized to produce lactic acid. Creatine phosphate is broken down to produce ATP for muscle contraction. So, both ADP and lactic acid increases in muscles.

(169) **Answer :** (3)
Solution:
Smooth muscle fibres are not striated. Smooth muscles lack the highly organised striated structure of actin and myosin filaments. Location of all the types of muscle fibres differ.
Skeletal muscle fibres are multi-nucleated.

(170) **Answer :** (4)
Solution:
The globular head of meromyosin is an active ATPase enzyme and has binding sites for ATP and active sites for actin.

(171) **Answer :** (3)
Hint:
'A' comprises 80 bones in humans.
Solution:
Axial skeleton includes cranium, vertebral column, ribcage and facial bones. Appendicular skeleton comprises limb bones and girdles.

(172) **Answer :** (4)
Hint:
Ball and socket joint
Solution:
Synovial joints help in locomotion and many other movements. Ball and socket joint (between humerus and pectoral girdle; hip joint), hinge joint (knee joint), pivot joint (between atlas and axis), gliding joint (between the carpals) and saddle joint (between carpal and metacarpal of thumb) are some examples of synovial joints.

(173) **Answer :** (3)
Hint:
Equal to 14
Solution:

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False ribs = 6 in number
Phalanges = 14 in each limb
Tarsals = 7 in each lower limb
Lumbar vertebrae = 5 in number
True ribs = 14 in number

(174) **Answer :** (1)
Solution:
Lower Jaw is made upof single bone Mandible.

(175) **Answer :** (3)
Hint:
Suture is included in this type of joint.
Solution:
First cervical vertebra (atlas) forms synovial joint. Its superior surface articulates with the occipital condyles to form an ellipsoidal joint. Its inferior surface articulates with axis vertebra to form a pivot joint. In between other cervical vertebrae, cartilaginous joints are present.

(176) **Answer :** (2)
Hint:
Clavicle is commonly called collar bone.
Solution:
Scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs. The dorsal flat triangular body of scapula has a slightly elevated ridge, called the spine, which projects as a flat expanded process called the acromion.
The clavicle articulates with the acromion. Below the acromion is a depression called the glenoid cavity which articulates with the head of the humerus.
At the point of fusion of the coxal bones, a cavity called acetabulum is formed, which articulates with the thigh bones.

(177) **Answer :** (3)
Solution:
At the point of fusion of ilium, ischium and pubis is a cavity called acetabulum to which the thigh bone articulates.
Glenoid cavity and acromion are associated with the pectoral girdle.
The two halves of the pelvic girdle meet ventrally to form the pubic symphysis containing fibrous cartilage.

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(178) **Answer :** (3)
Solution:
Femur is the longest bone in the human body.
Sternum is a flat bone on the ventral midline of thorax.
A cup-shaped bone, called patella, covers the knee ventrally (knee cap).

(179) **Answer :** (2)
Solution:
False ribs (floating ribs) are also attached dorsally to thoracic vertebrae.

(180) **Answer :** (2)
Solution:
Muscle fibre is the anatomical unit of muscle. Each muscle fibre has many parallelly arranged myofibrils. Each myofibril contains many serially arranged units called sarcomere which are the functional units.
Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles.

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